

DeepLearning.AI

Evaluating & Improving Agentic Applications

Laurie Voss

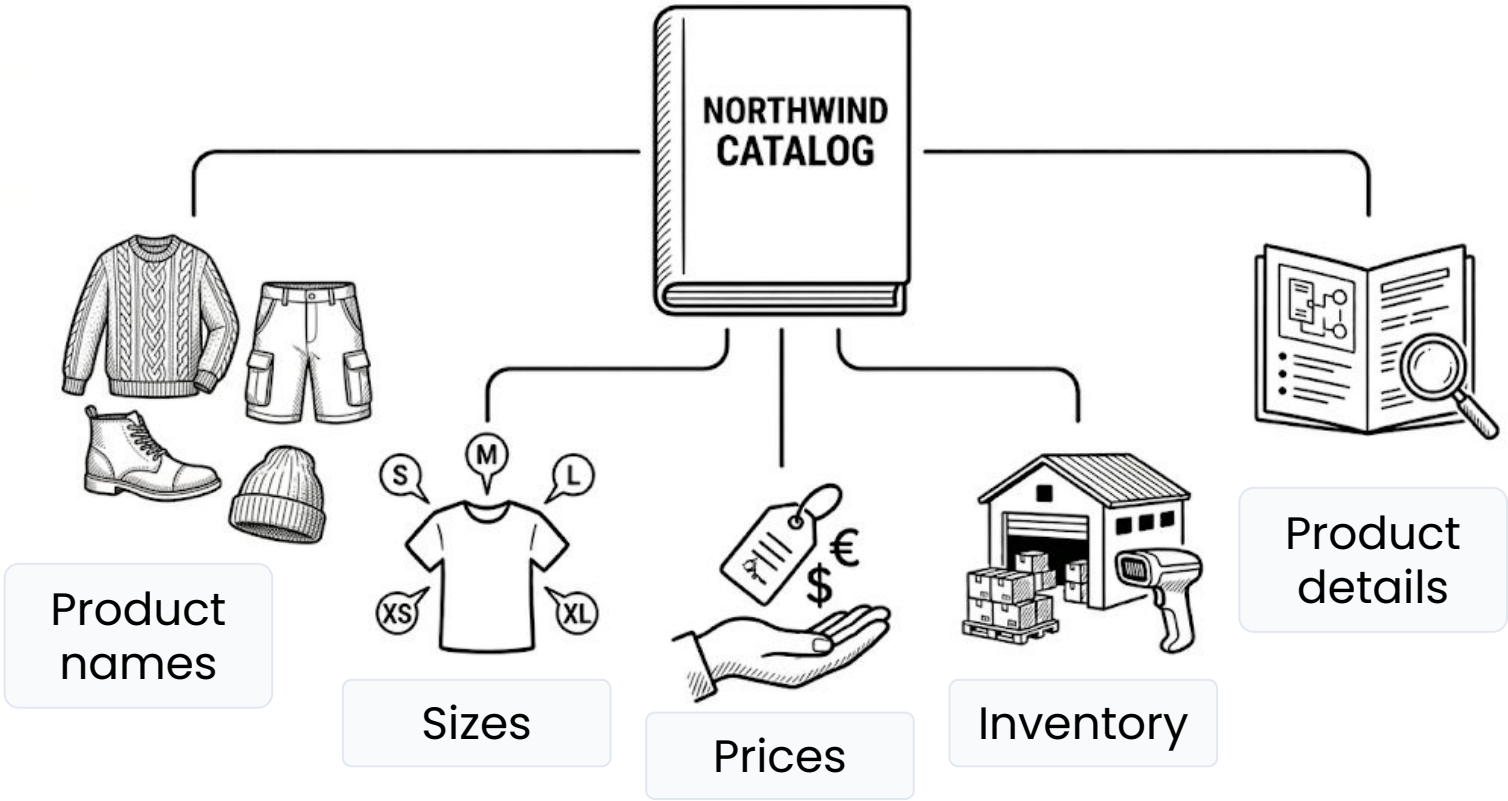


DeepLearning.AI

Why evals matter

Module 1 · Video 1

Meet Northwind Apparel



Meet the Northwind support agent

Find suitable products

↓

Check stock

↓

Add to cart



Some tasks follow rules, others need judgment

Rule in code → Deterministic



Wool sweater, size L

Stock: 0

Out of stock

Recommendation → Requires judgment



First winter camping trip



Fleece pullover

Holds heat at night



Hoodie

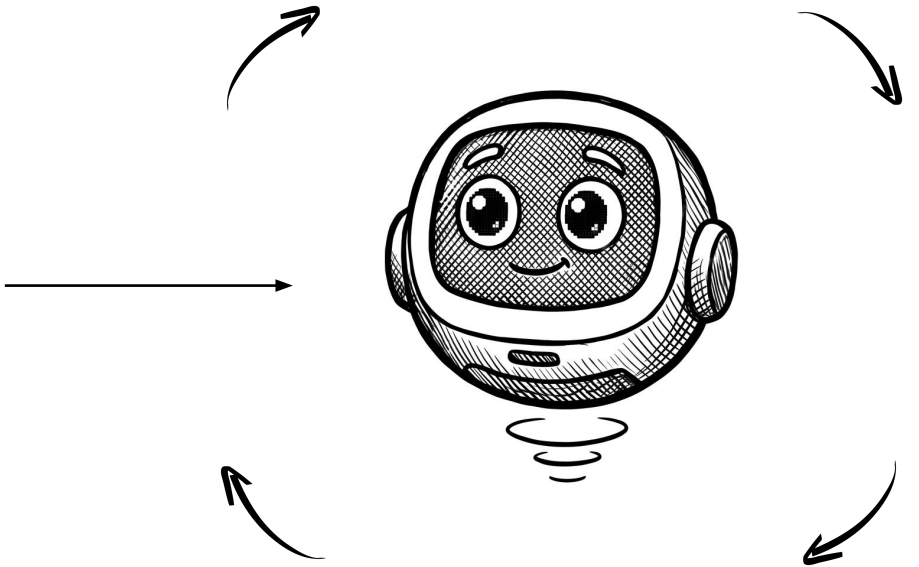
Easy to layer

An agent works toward a goal

Goal: a winter camping outfit

Instructions from Northwind

Tools: search catalog · check stock · add to cart



More you can build, harder to trust

A few new abilities

- Understand requests
- Make judgments
- Use tools toward a goal

A huge range of possibilities

- Support agents
- Coding agents
- Research tools
- Tutoring apps
- Travel planners
- Data analysis

But can you trust what you've built?

The same request can produce different decisions



Customer
First winter camping trip



Fleece pullover
Cold-weather layer



Cargo shorts
Summer wear

Request

What should I wear on my first winter camping trip?

Run 1

A fleece pullover will keep you warm at camp.

Run 2

Cargo shorts are a great pick for this trip!

One passing run doesn't make it reliable

A reasonable answer can hide a bad decision

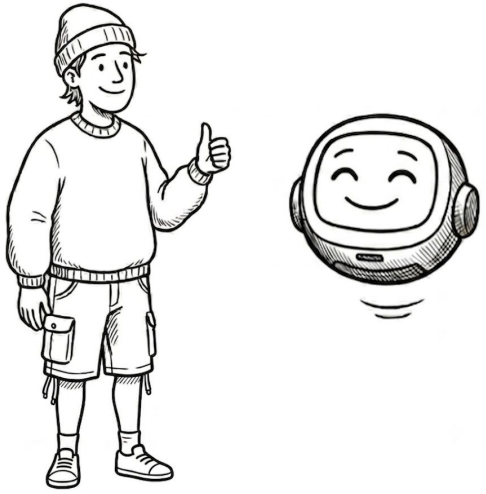
What should I wear on my first winter camping trip?



Done! I added a sweater, cargo shorts, and a beanie to your cart.

Cart updated

These cargo shorts will **keep you warm** on your winter trip.



Good answer, wrong decision

The result doesn't show what the agent did

What you see



Done! I added a sweater, cargo shorts, and a beanie to your cart.

Cart updated

What the agent did

Search catalog → Check product details → Recommend products → Add to cart

Skipped

People ask in ways you can't anticipate

Said directly

What should I wear for **winter** camping?



A fleece, wool socks, and a beanie will keep you warm.

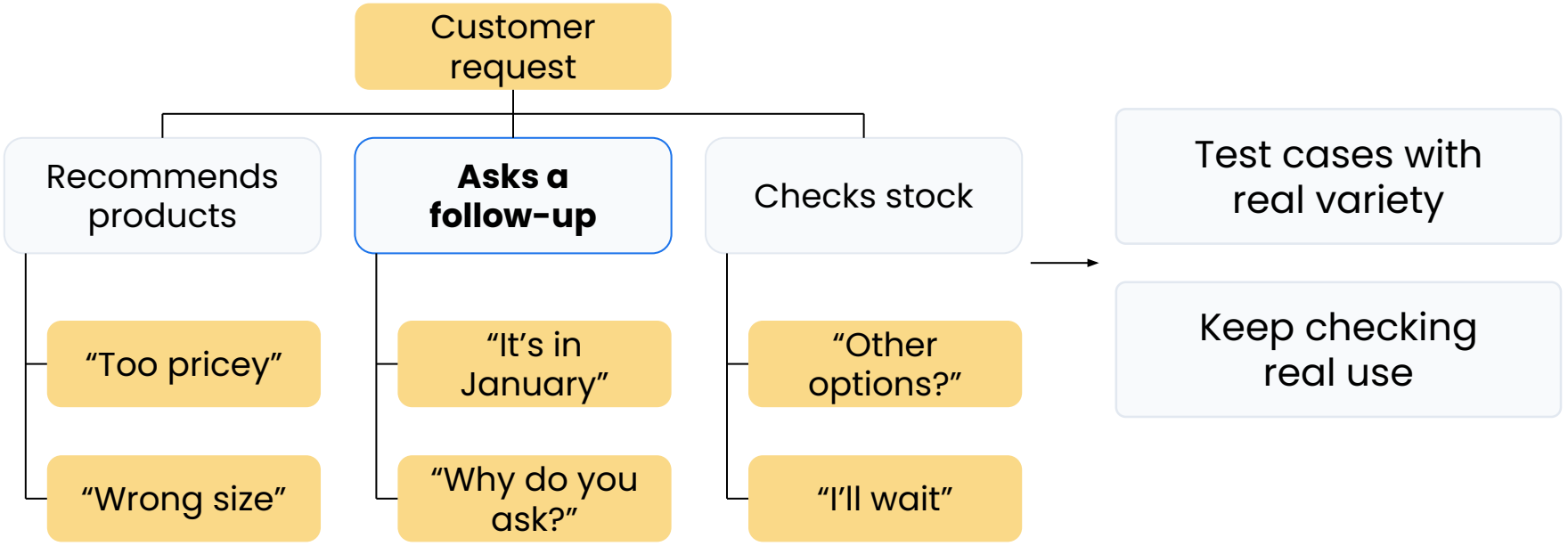
Buried in a longer description

My friend invited me on a trip to the mountains. We'll hike to a lake and camp for two nights. **It's winter**, so we'll cook over a fire. What should I buy in size large?



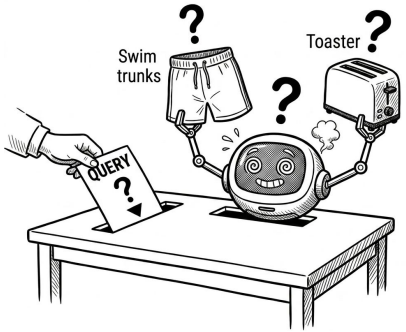
Here are some picks: a T-shirt, cargo shorts, and sandals.

Every reply opens new paths



Three reasons AI apps are harder to test

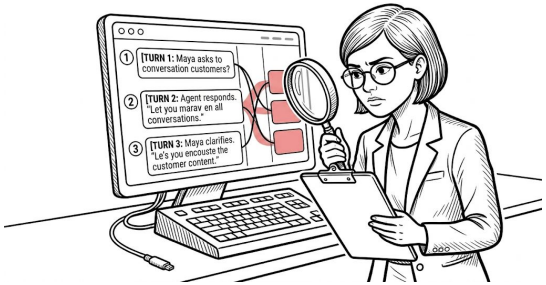
Behavior changes run to run



Mistakes can look like success



Too many situations to cover



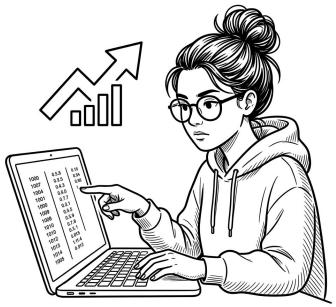
Checking only code and tools misses these

Systematic evaluation gives you evidence

~~"I think it's working"~~

"I can prove it's working"



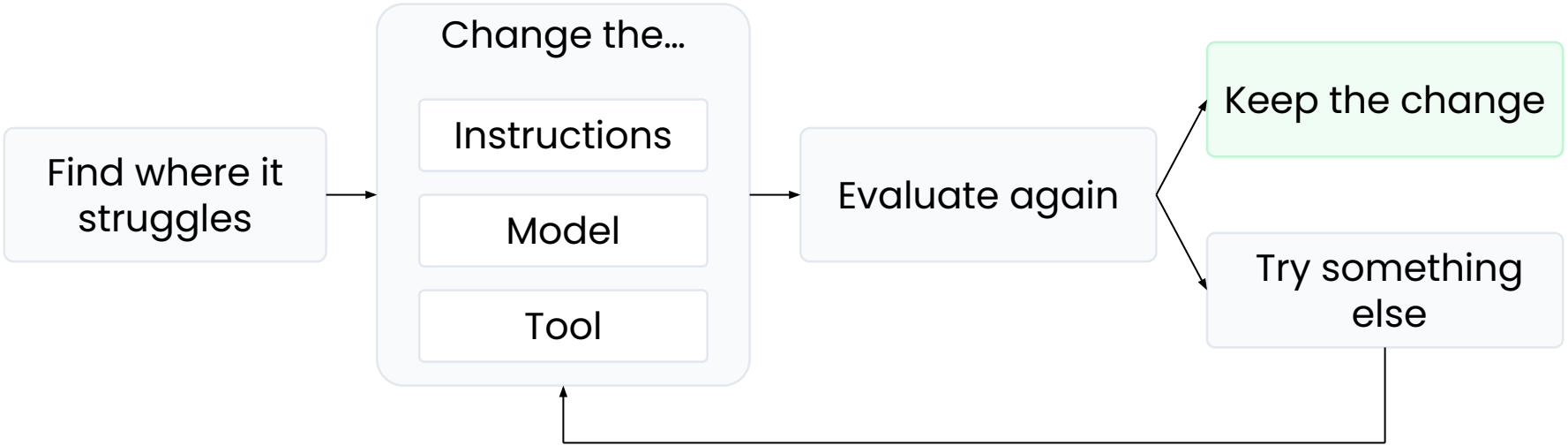


Against what the task requires

Realistic situations

Repeated runs

Evaluate before and after every change



1 Look closely at a few examples

2 Evaluate systematically as you scale



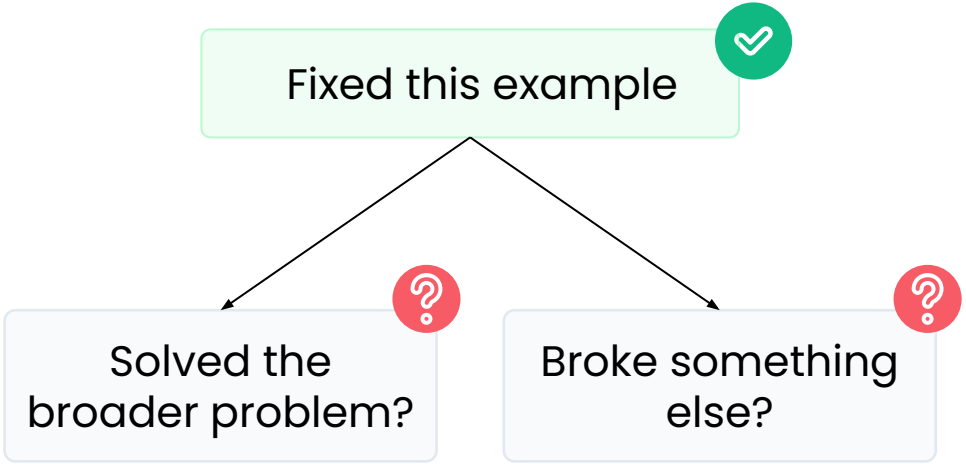
DeepLearning.AI

The Evals Process Part 1

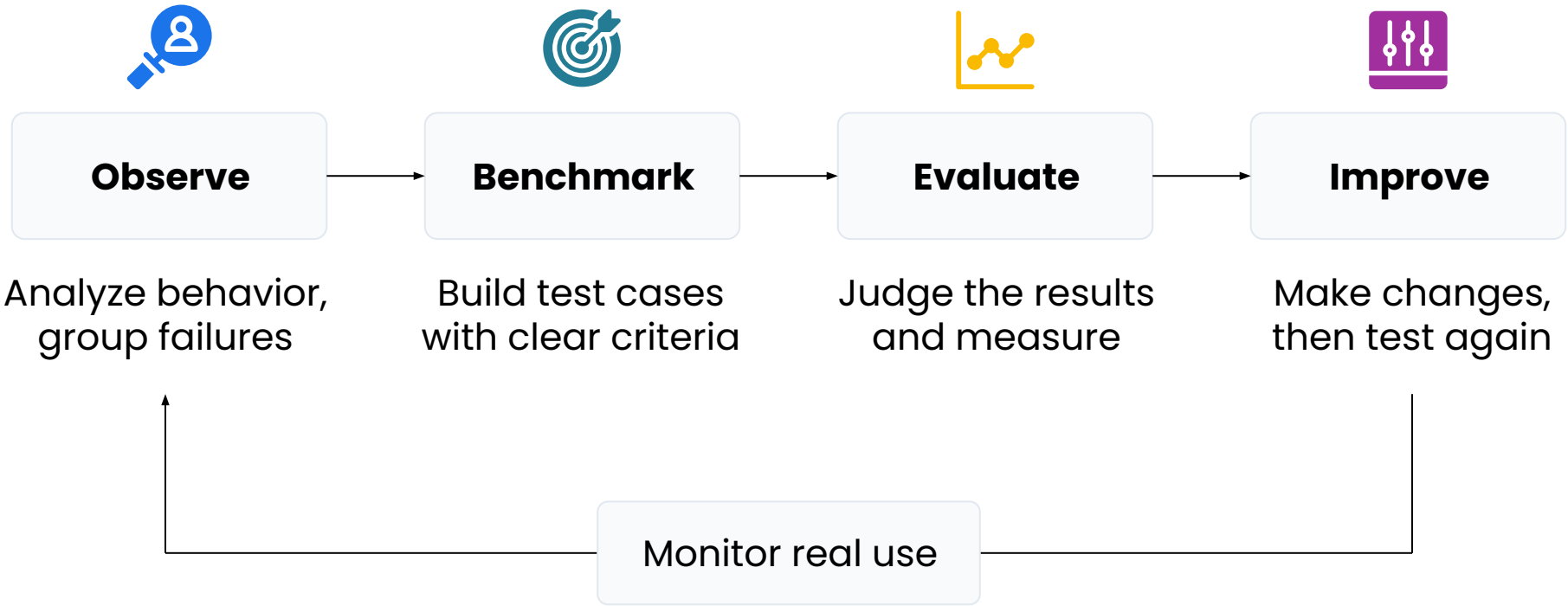
Module 1 · Video 2

Fixing one failure at a time leaves you guessing

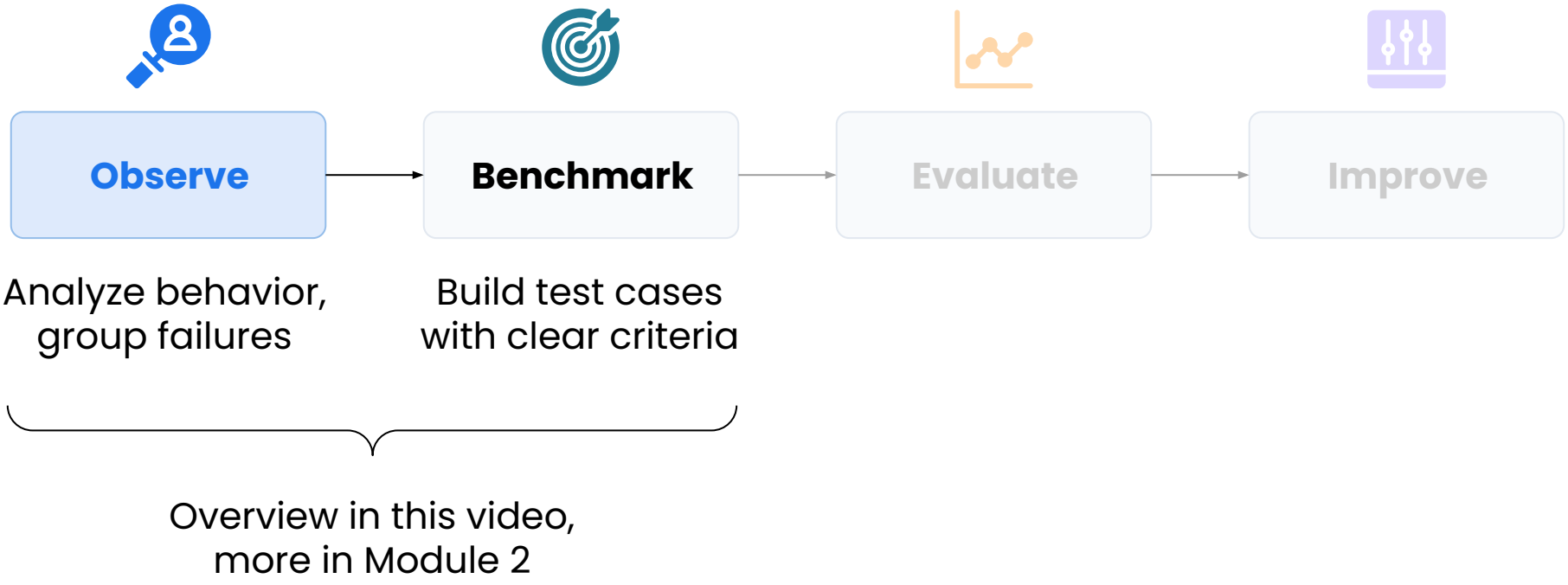




The evaluation process in four parts



Parts 1 and 2: Investigate and Define



A failure is a comparison



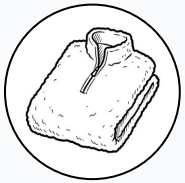
What should I wear on my first winter camping trip?

What the agent did



Cargo shorts
Summer wear

What it should have done



Fleece pullover
Cold-weather layer

Requirements: what to accomplish · what to get right · what to avoid

You can’t judge what you haven’t decided



I need a warm jacket for winter camping.

No size given

Ask first



What size do you wear?

PASS

FAIL

Suggest first



Here are three warm jackets you might like.

PASS

FAIL

Where requirements come from



Only recommend items that are in stock





Customer needs

Company policies

People who know the work

Requirements

Make your expectations explicit

Task: recommend clothes

More than one good answer



Fleece pullover



Hoodie

Requirements

- Suits the customer's needs
- Reflects the catalog and stock
- + Clarify and add as you learn

Start with error analysis

Observe → Benchmark → Evaluate → Improve



Look closely at a few examples



Compare with requirements



Group into patterns

Two kinds of problems to tell apart

Are the instructions clear and complete?

Instructions: Recommend warm clothes that suit the customer's trip.

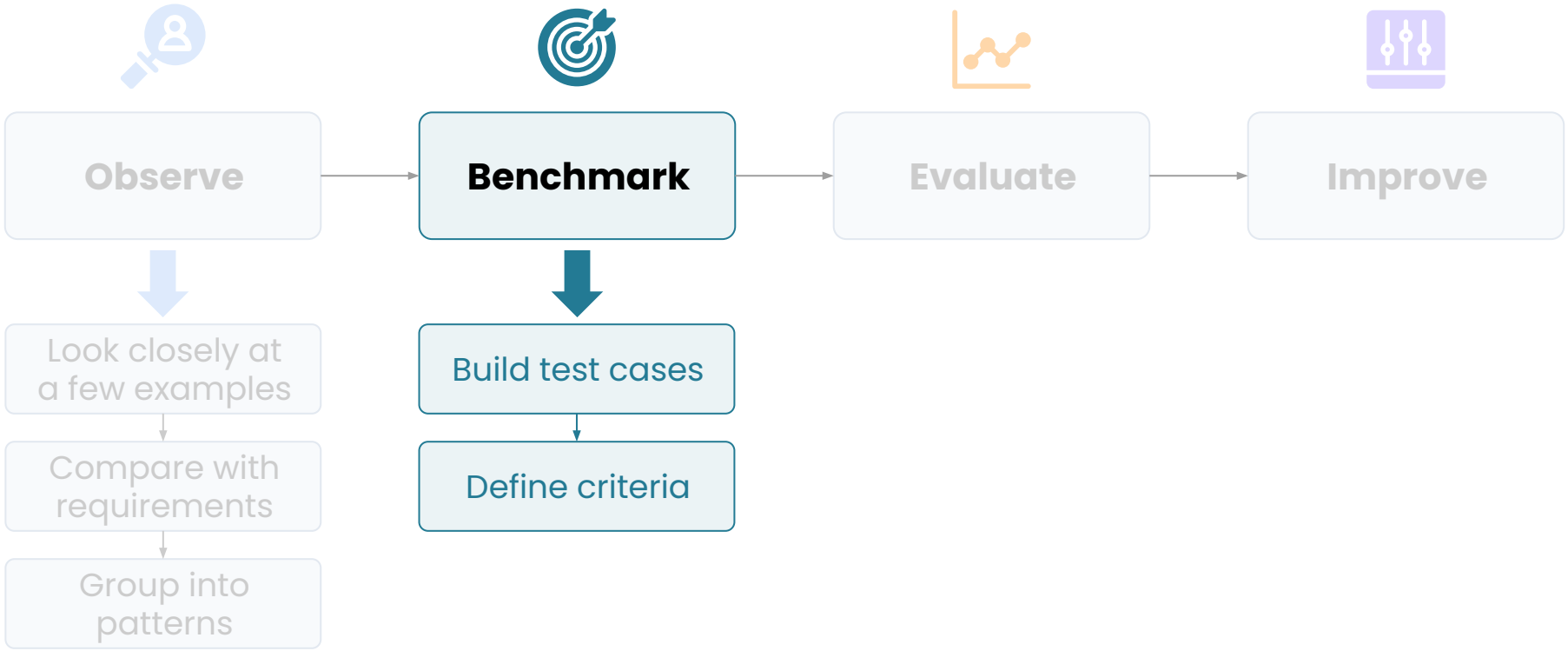
Missing: check the size

Can the model follow them reliably?

Instructions: Recommend clothes that suit the trip, in sizes that are in stock.

Suggests a sold-out jacket

Part 2: Define



Each case pairs a situation with what you expect

Customer query	Context	Expected outcome
What should I wear on my first winter camping trip?	Fleece and hoodie in stock	**Recommend warm, in-stock layers**
I need a warm jacket for winter camping.	No size given	**Ask for the size first**
Do you have the wool sweater in size L?	Stock: 0	**Say it's out of stock**

Evaluation set

Test more than past failures



Past failures

Cargo shorts
for a winter trip



Everyday work

A fleece for
a cold night



Challenging situations

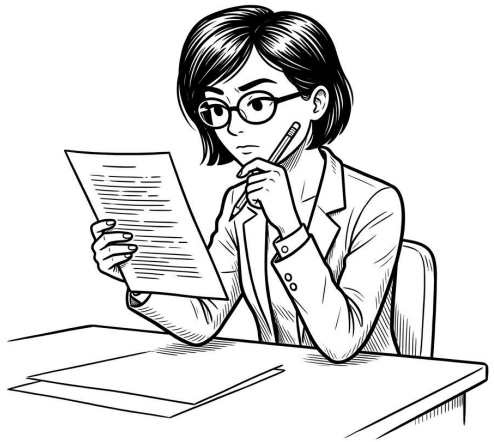
A long request,
no size given

Reuse after every change · Update as you learn

A rubric is a written guide

Rubric

- What to check in each result
- What counts as pass or fail
- Information needed to decide
- Pass/fail examples, explained



Check each requirement separately



Cargo shorts, size L

Availability

Size L is in stock

Suitability

Not warm enough

Try the rubric on real results



I need a warm jacket for winter camping.



This jacket will work well for your trip.

Suits the customer's needs

PASS

FAIL

You may need more information or a clearer requirement

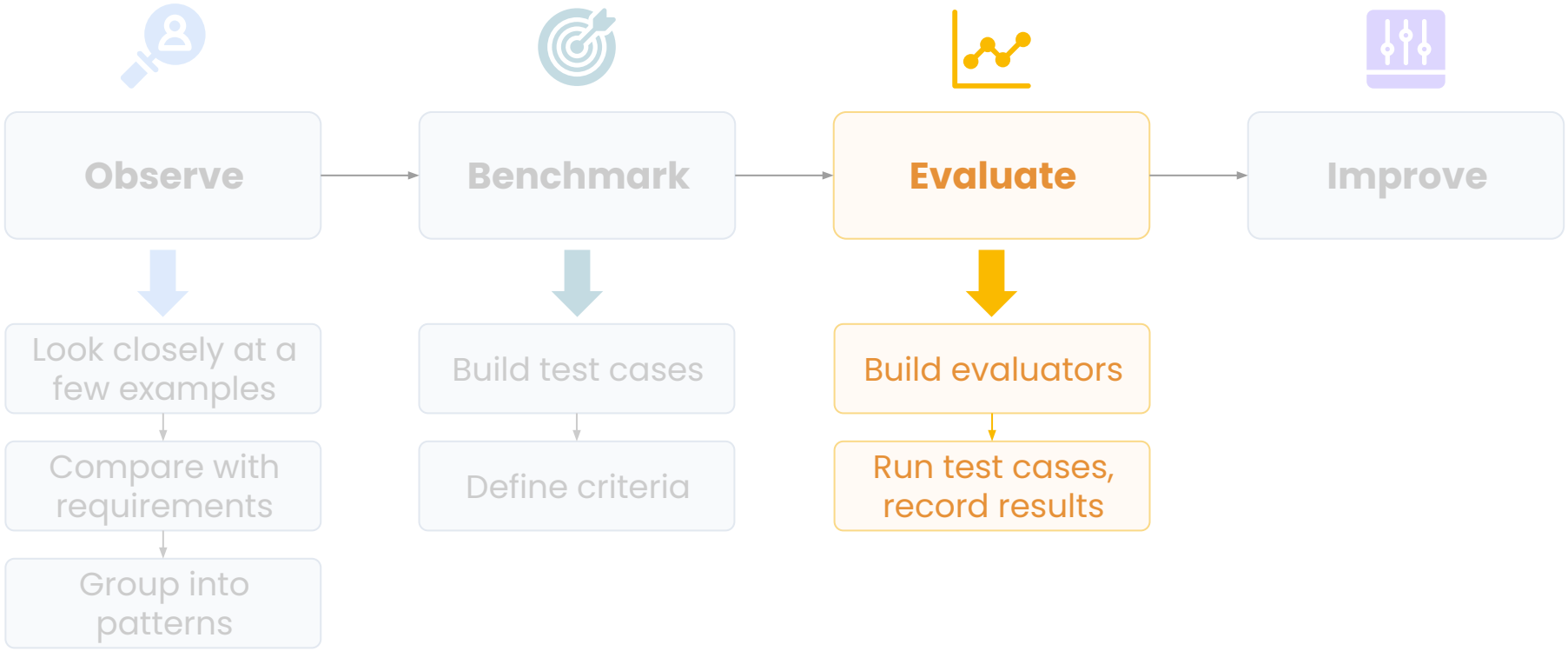


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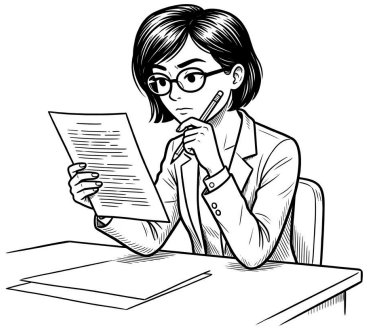
The Evals Process Part 2

Module 1 · Video 3

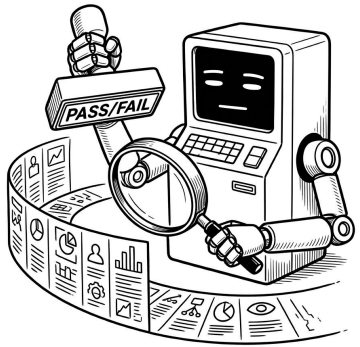
Part 3: Evaluate



Three ways to judge the results







Human-based	Code-based	LLM judge
Is this a good answer for this customer?	Is the item in stock in the requested size?	Do these clothes suit a winter camping trip?

Build evaluators in code or on a platform



Your own Python code

Both kinds of evaluator,
no platform needed



An evaluation platform

Runs the checks and
organizes the results

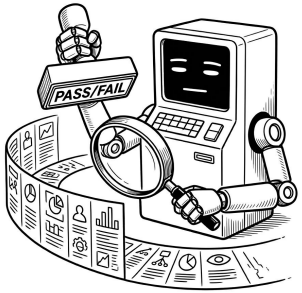


Phoenix · open source · from Arize

Check the evaluator against people



Same result, same criteria



Reviewer
A person

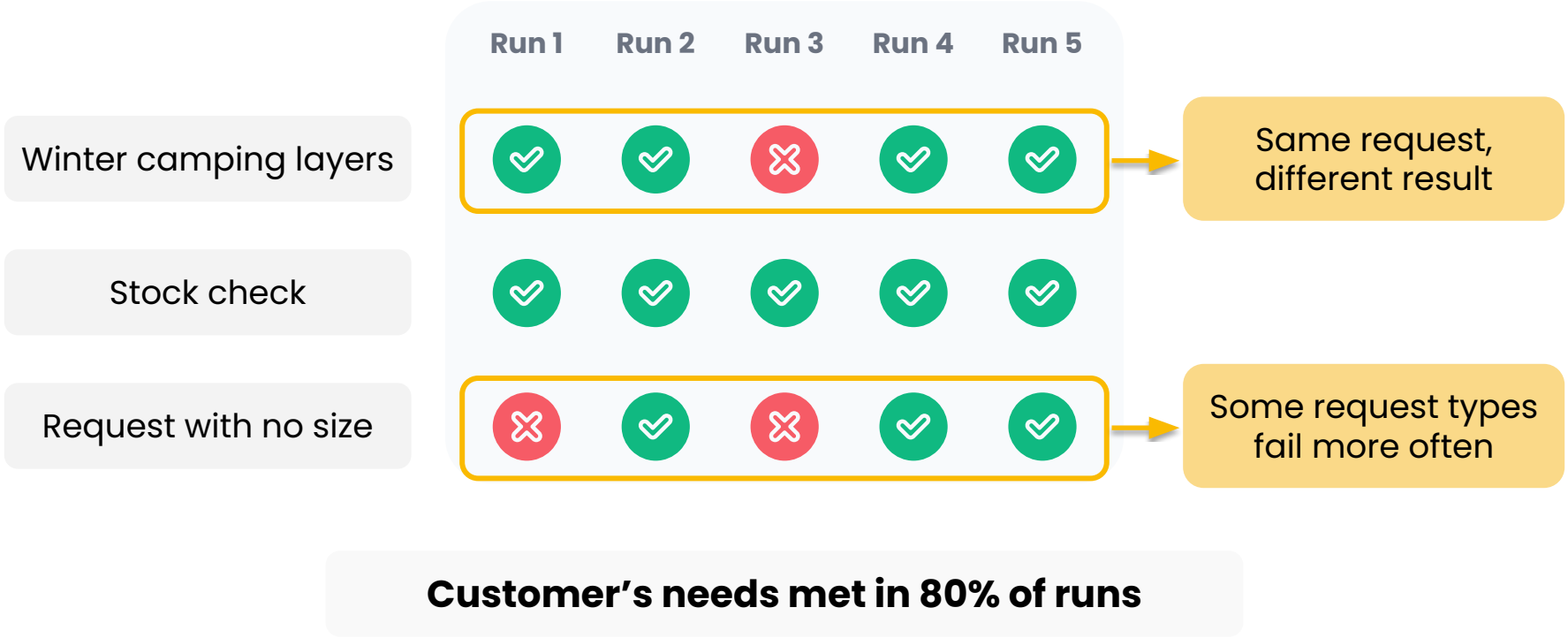
PASS

Disagree? Check the evaluator, the reviewer or the criteria

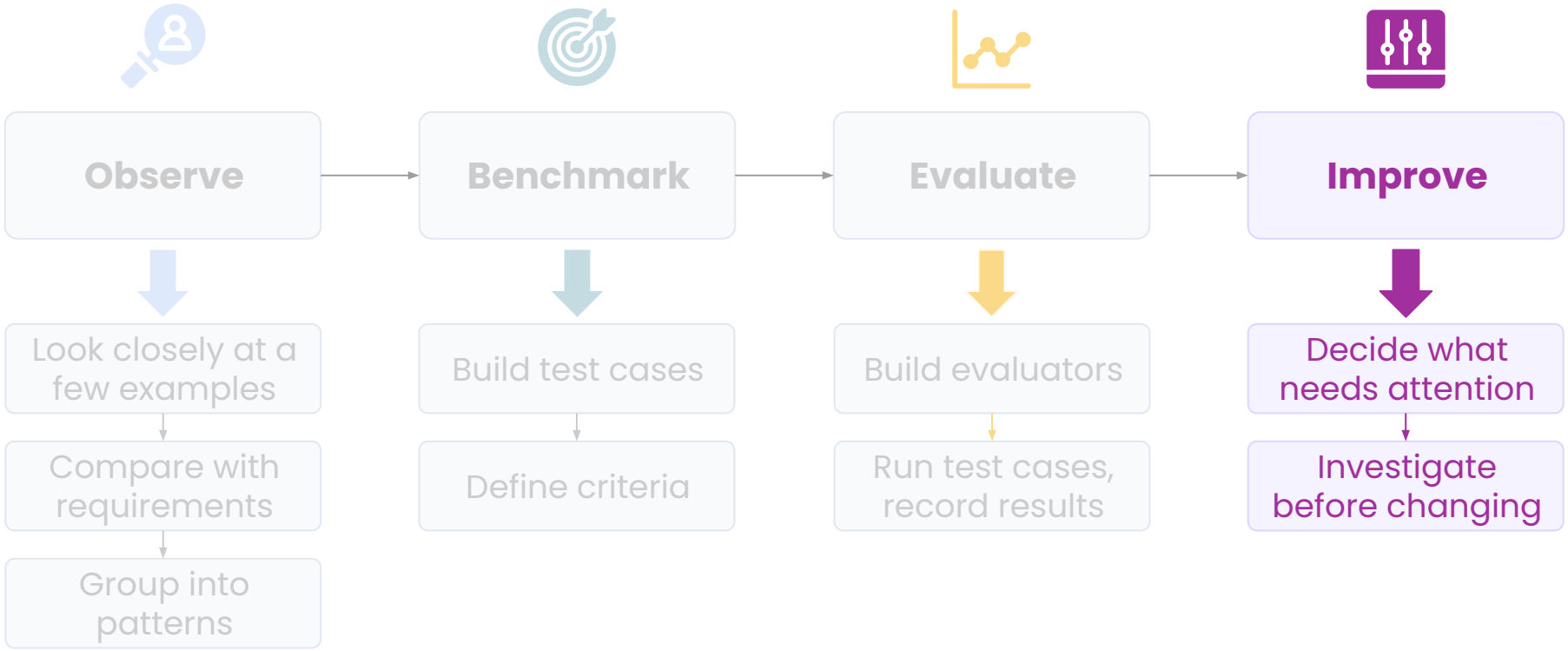
Evaluator
Code or LLM judge

FAIL

Measure across cases and repeated runs



Part 4: Optimize

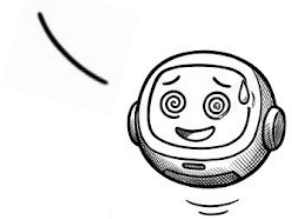


Match the fix to what went wrong

What the agent said

Here is what to keep in mind:

- Machine washable on wool cycle
- Cool-water machine wash
- Wool-specific detergent
- Cedar blocks for moths



What the tool returned



Catalog

Merino wool sweater. Machine washable on wool cycle

The agent's instructions

```
Use tools when helpful.
Give concise answers.

Never state a product
detail unless a tool
result supports it.
```

Other fixes to try

Better information
Tools and workflow
A different model

Change the agent, but keep the evaluation fixed

New instruction

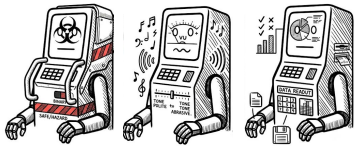
Agent v1



Test cases



Evaluators



Baseline results

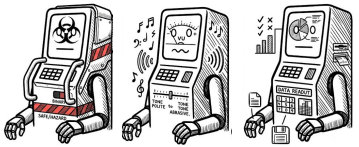
Agent v2



Test cases



Evaluators



New results

Catch what your change breaks

Did the target improve?

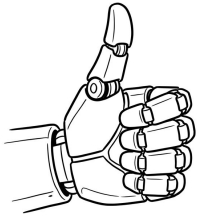
- Merino wool care ✓
- Hiking shoe materials ✓
- Denim jacket care ✓

Still doing what it did well?

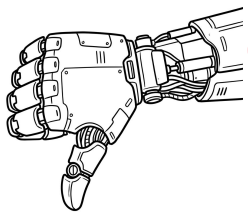
- Jeans size availability ✗
- New season items ✓
- Basic shirt colors ✓

Regression: losing behavior that worked before

Improvement can introduce tradeoffs



- Less unsupported advice
- Target failure fixed
- Catalog-backed answers

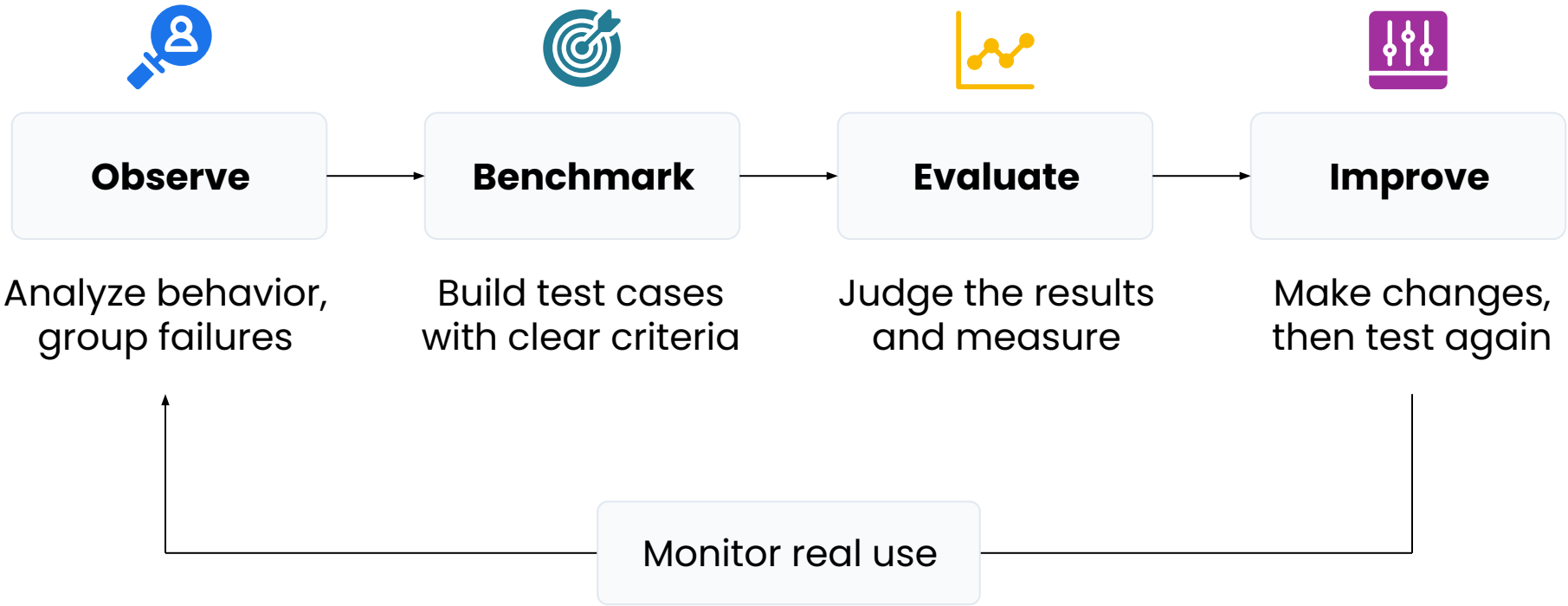


- Unnecessary tool calls
- Slower responses
- Higher cost

Keep | Revise | Try something else

Better than before doesn't mean good enough

The evaluation cycle



The output doesn’t show what the agent did

What you see

What should I wear on my first winter camping trip?

A fleece pullover will keep you warm at camp.

What the output doesn’t show

Which information did it use?

Which tools did it call?

Did it skip something important?

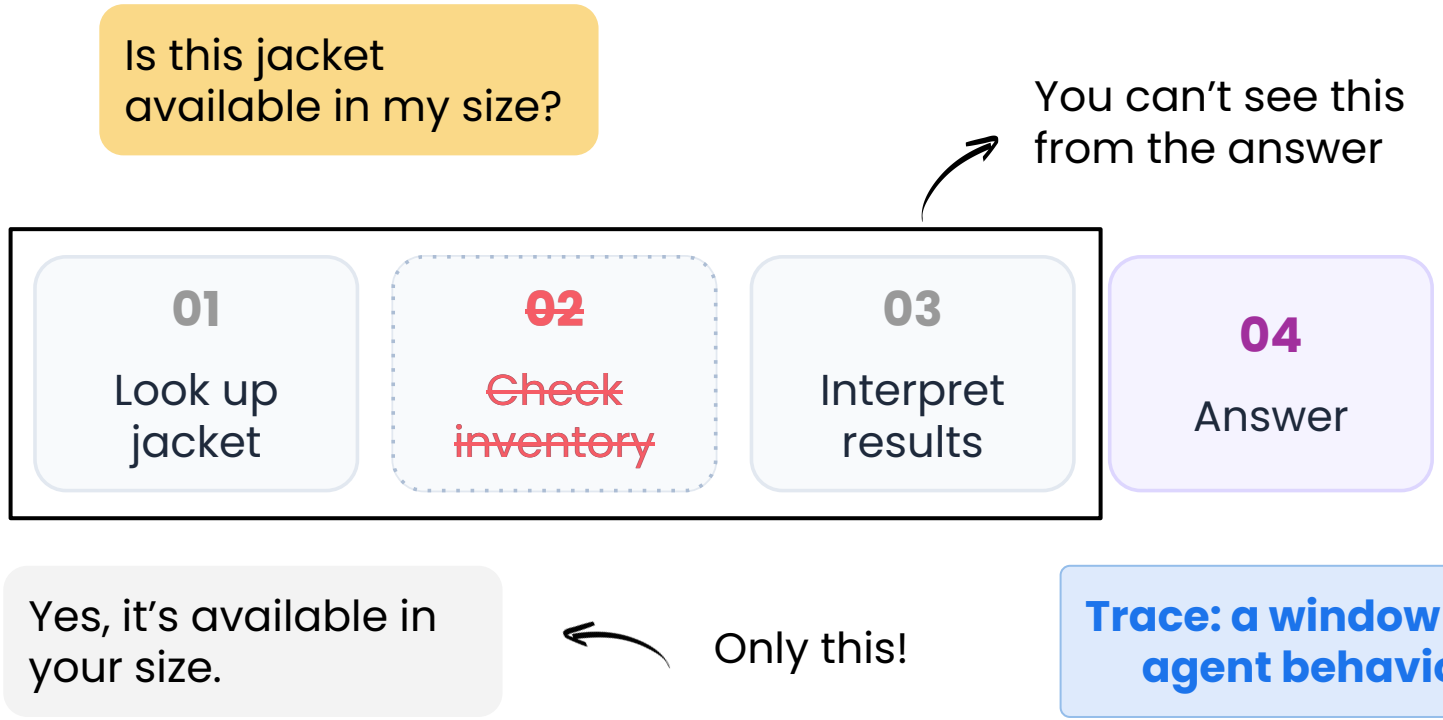


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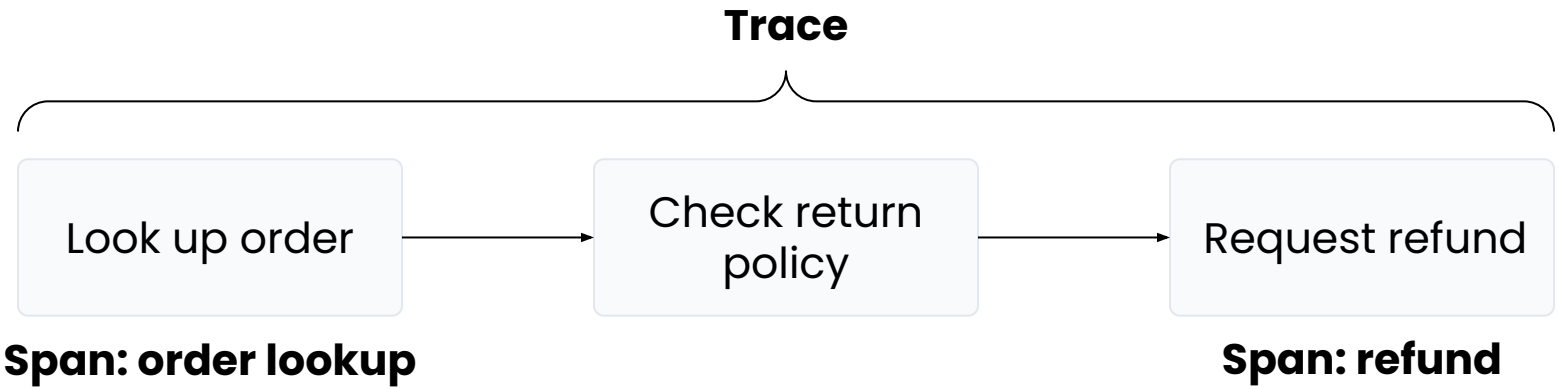
Traces

Module 1 · Video 4

Input and output hide the steps between

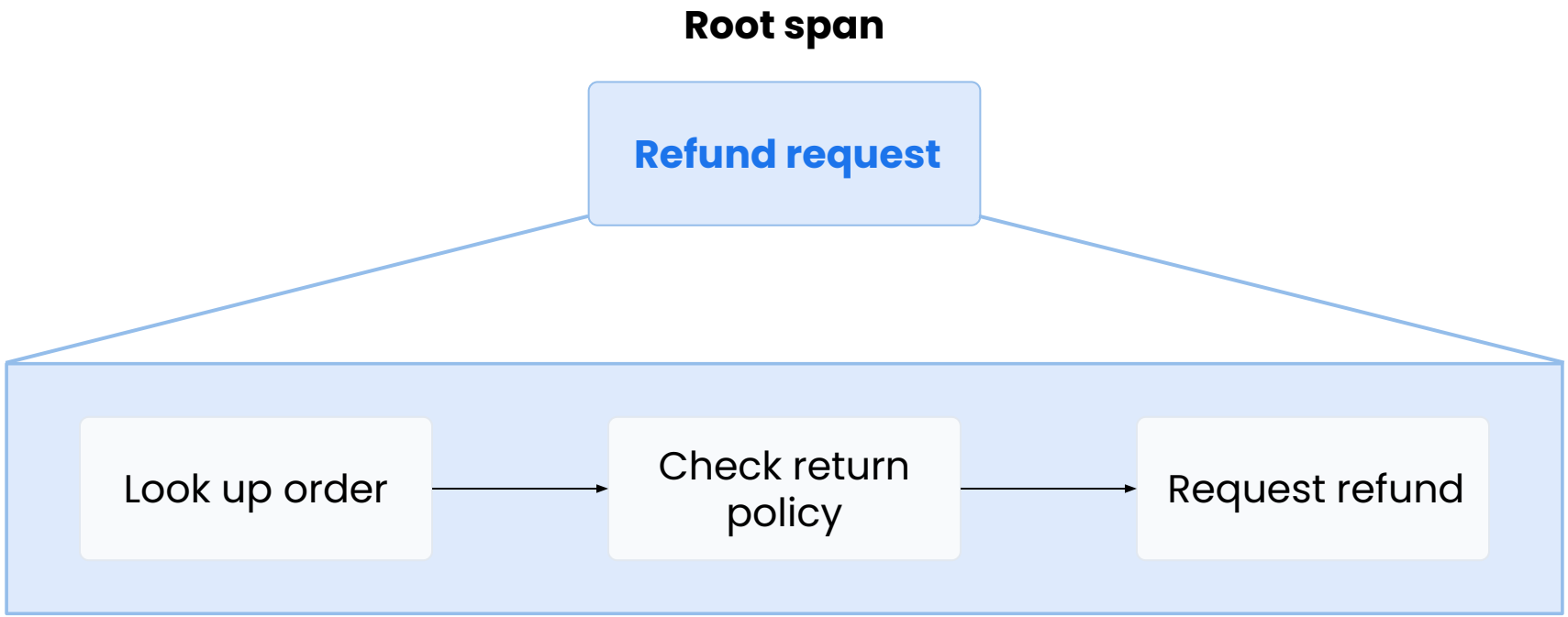


Each step in a trace is recorded as a span



A span is one recorded operation, with a beginning and end

A trace shows how the spans fit together



What a tool-call span records

Look up order

```
{
  "name": "lookup_order",
  "span_type": "tool",

  "input": {
    "order_id": "A99"
  },

  "output": {
    "item": "Wool jacket",
    "quantity": 1,
    "purchase_price": 89.00
  }
}
```

Which operation ran, and its type

What was sent to the tool

What came back

What a model-call span records

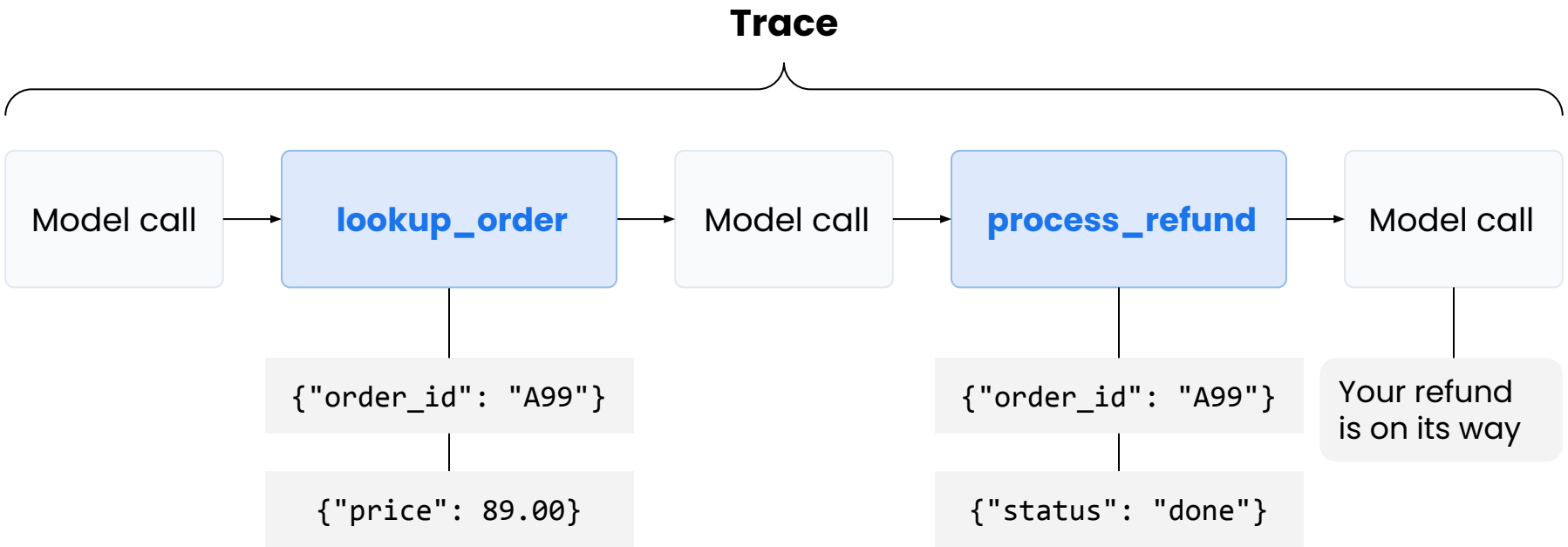
Model call


```
{
  "input": {
    "customer_message": "I want to return my jacket",
    "instructions": "You are Spark, a support agent",
    "tool_results": "Order A99: wool jacket, $89.00"
  },

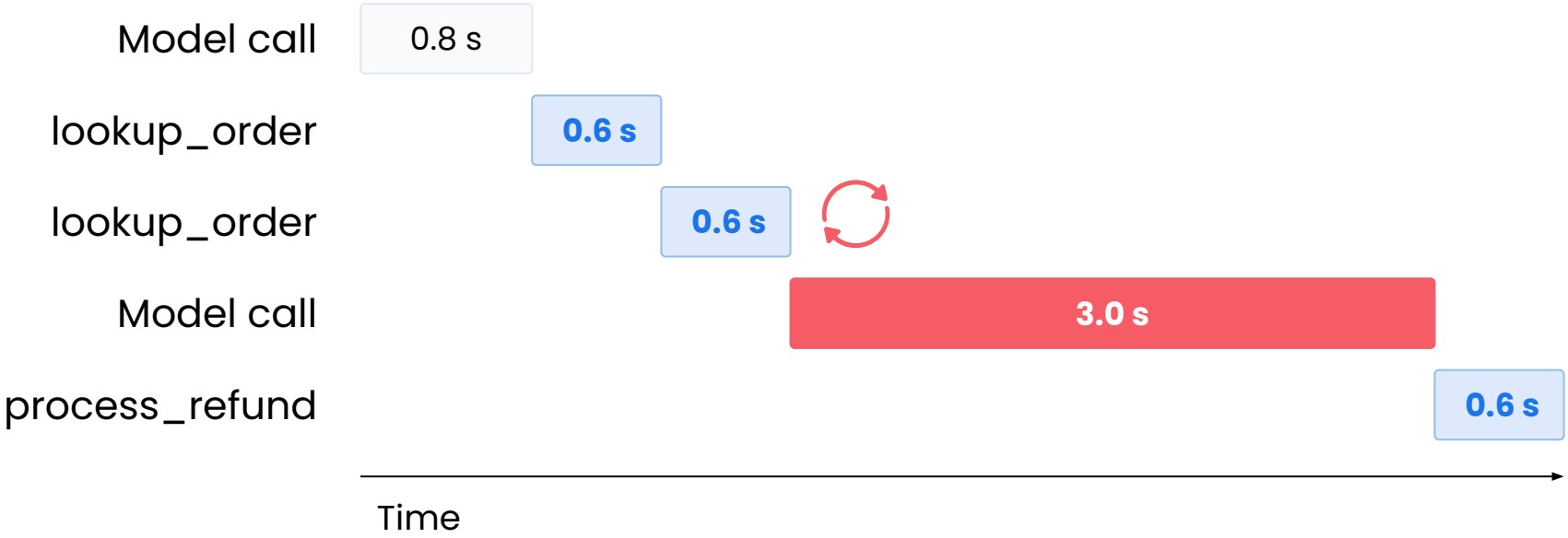
  "output": {
    "type": "tool_request",
    "tool": "process_refund"
  },

  "attributes": {
    "model_version": "v2.1",
    "prompt_version": 3,
    "total_tokens": 412
  }
}
```

A trace connects the steps in a run



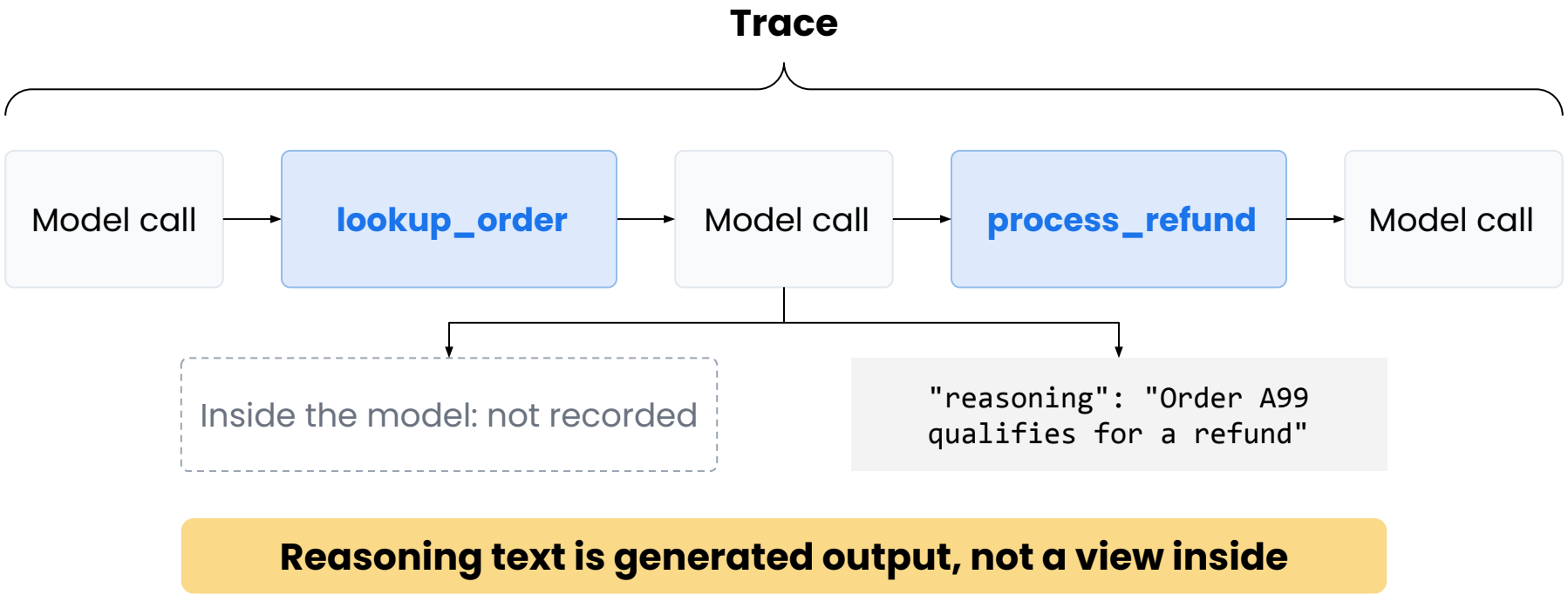
Spans record timing



Model call or order lookup?

Correct answer, unnecessary work

A trace doesn't show the model's internals



Reasoning text is generated output, not a view inside

Instrumentation records the spans

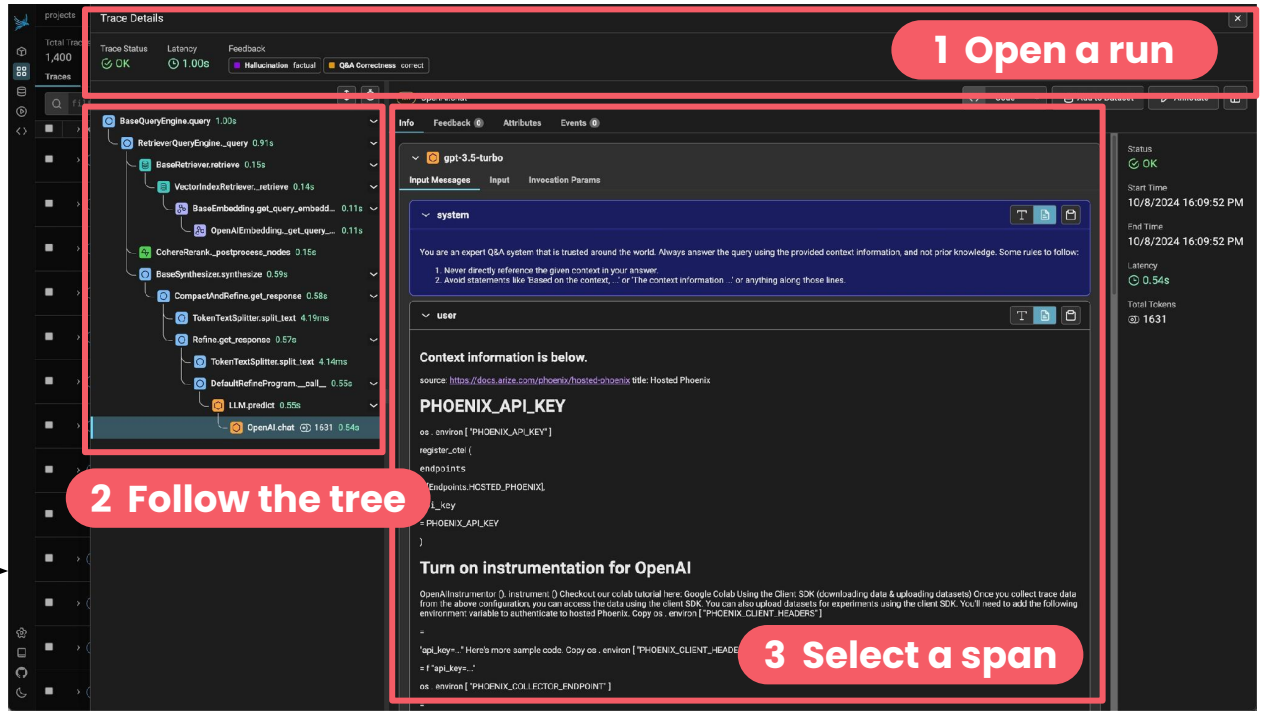
Source of the spans	What gets recorded
Tracing integration	Captures model calls and tool calls: inputs, outputs, timing
Your own functions	You add recording around code the integration doesn't cover

Inspect traces in Phoenix

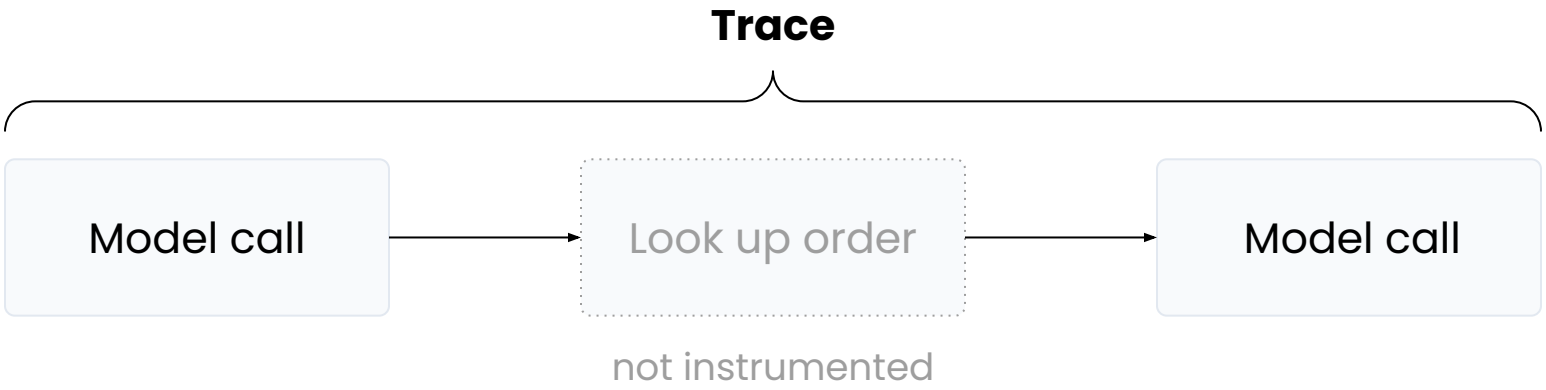
Your agent

Tracing integration

Phoenix

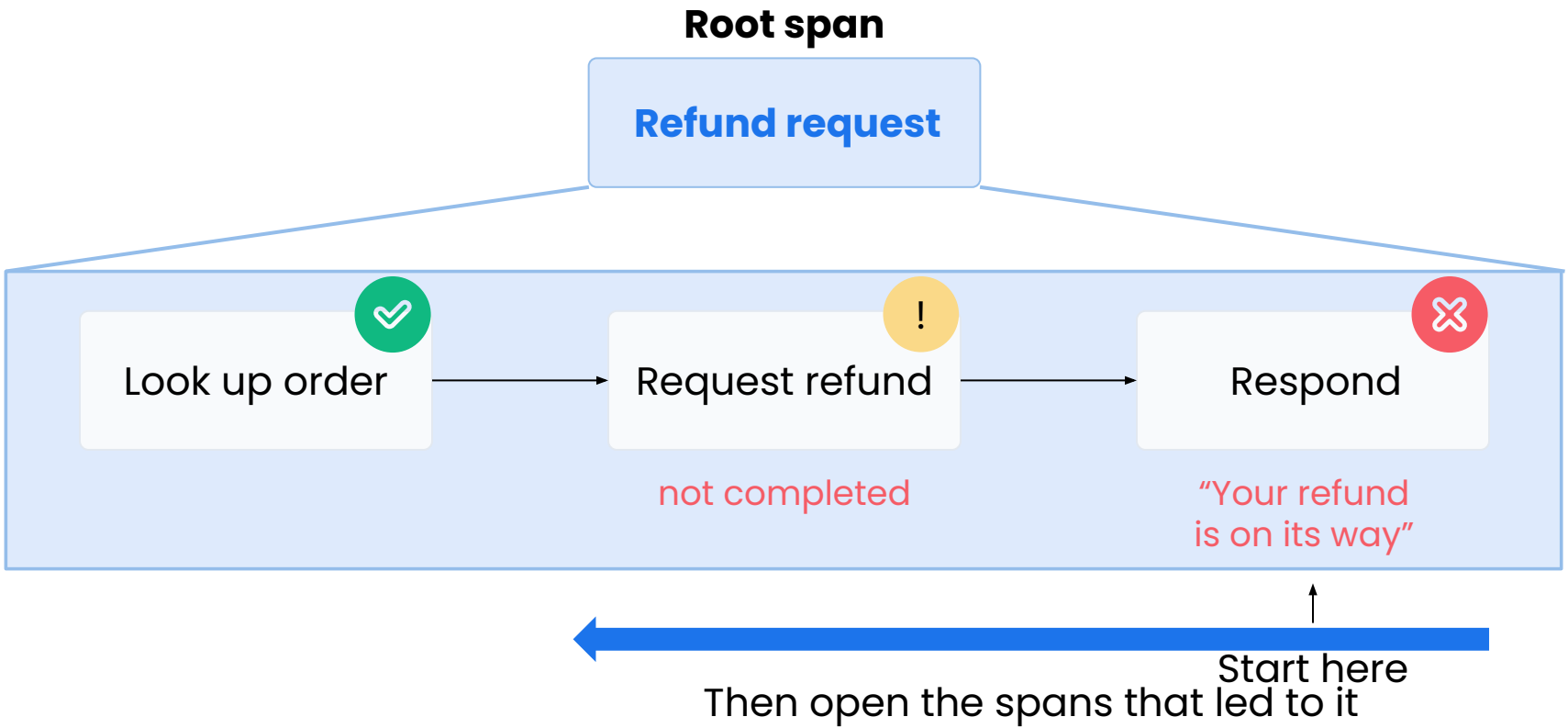


A missing span isn’t a skipped step

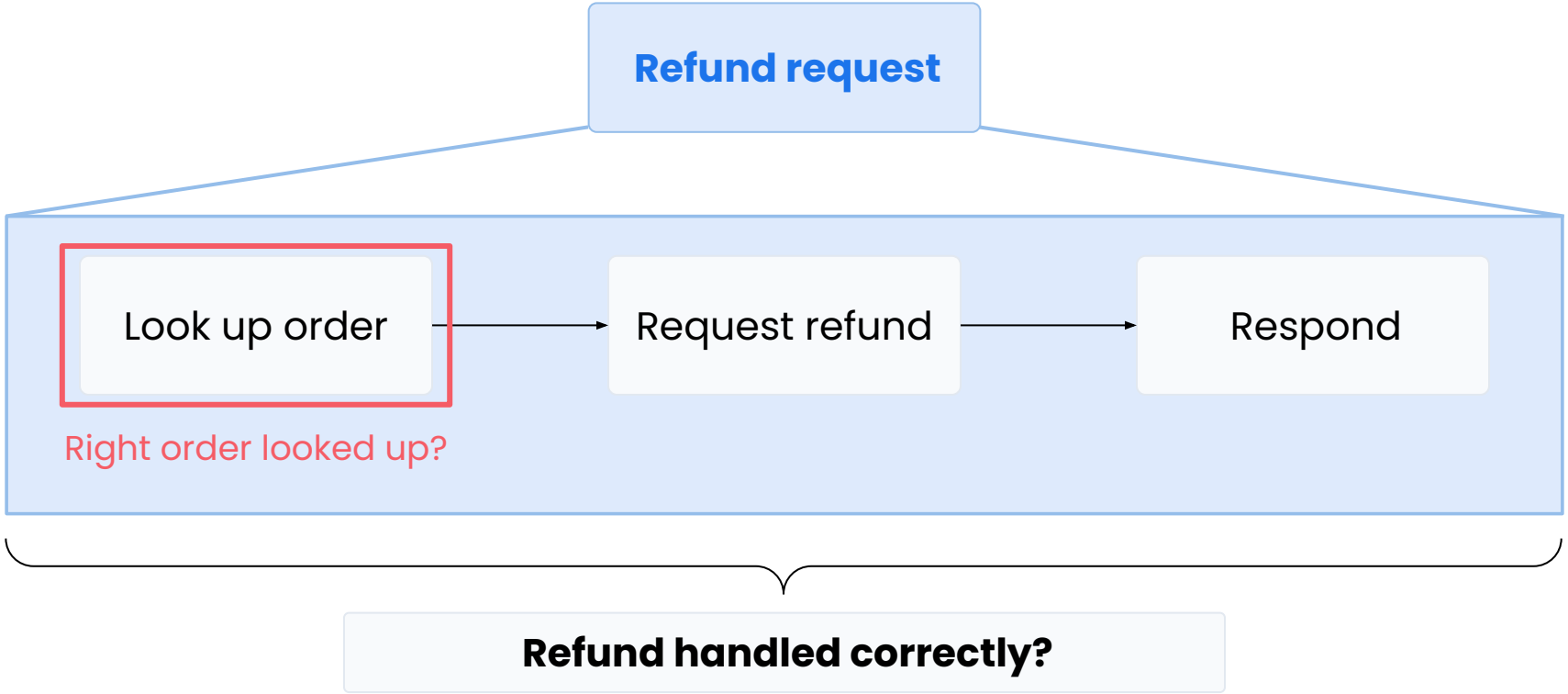


Absent span ≠ skipped action

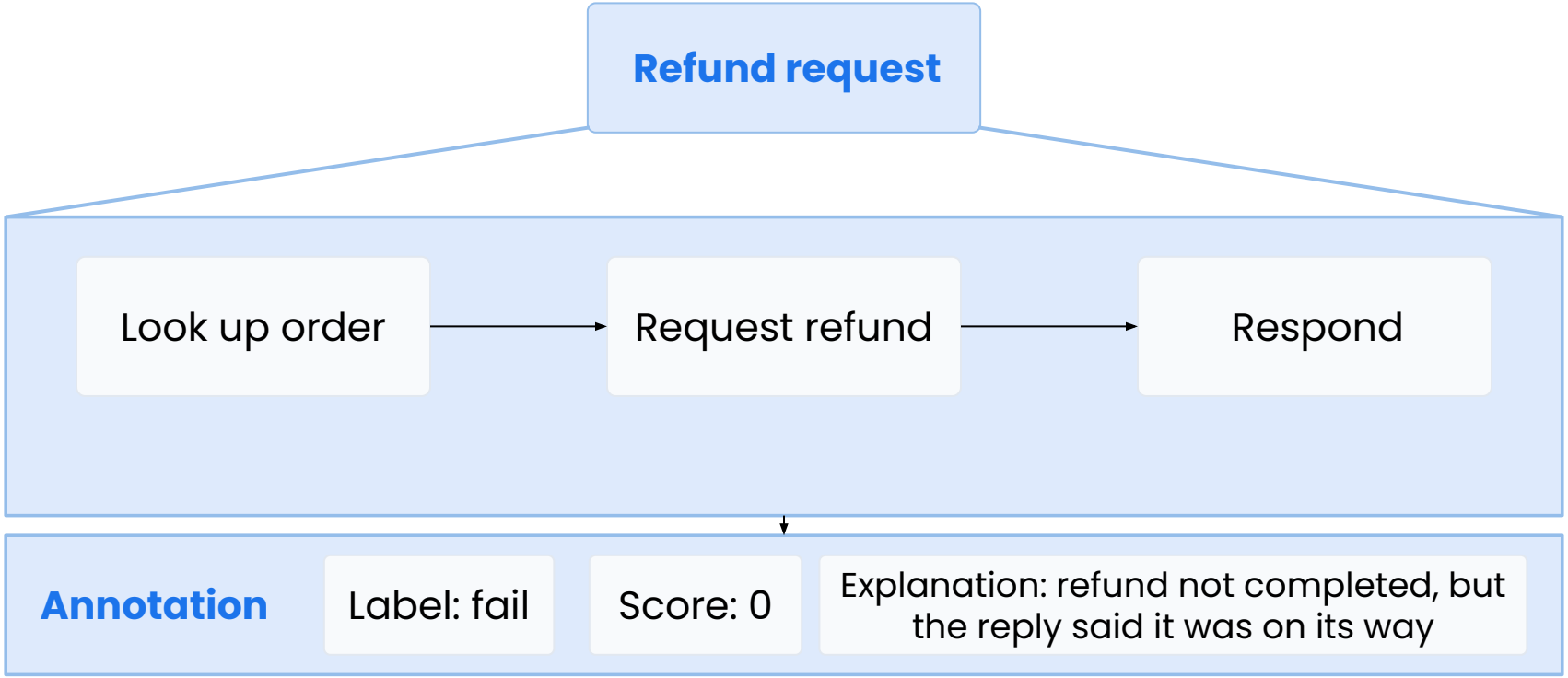
Follow the trace to where the decision went wrong



Evaluate one operation or the whole run



Record your assessment as an annotation



Trace vs. session

I'd like a refund for order A99

```
{
 "trace_id": "t1",
 "session_id": "s42",
 "spans": 4,
 "output": "Refund requested"
}
```

How long will the refund take?

```
{
 "trace_id": "t2",
 "session_id": "s42",
 "spans": 2,
 "output": "5 business days"
}
```

